

Screenshot 2025-07-19 093717

Step-1: First the registers are initialized with corresponding values (Q = Dividend.
 M = Divisor, $A = 0$, n = number of bits [n dividend])

Step-2: Check the sign bit of register A

Step-3: If it is 1 shift left content of AQ and perform $A = A + M$, otherwise shift left AQ
'and perform $A = A - M$ {means add 2's complement of M to A and store it to A}

Step-4: Again the sign bit of register A

Step-5: If sign bit is 1, $Q[0]$ become 0 otherwise $Q[0]$ become 1 ($Q[0]$ means least significant bit of register Q)

'Step-6: Decrements value of N by 1

Step-7: If N is not equal to 0 go to Step 2 otherwise

'go to next step

Step-8: If sign bit of A is 1 then perform $A = A * M$

Step-8: Register Q contains quotient and A contains remainder